

Single-Cycle RISC-V Reference

1. Instruction Formats

Type	[31:25]	[24:20]	[19:15]	[14:12]	[11:7]	[6:0]
R-type	funct7	rs2	rs1	funct3	rd	opcode
I-type	imm[11:0]	--	rs1	funct3	rd	opcode
S-type	imm[11:5]	rs2	rs1	funct3	imm[4:0]	opcode
SB-type	imm[12,10:5]	rs2	rs1	funct3	imm[4:1,11]	opcode
U-type	imm[31:12]	--	--	--	rd	opcode
UJ-type	imm[20,10:1,11,19:12]	--	--	--	rd	opcode

2. Control Signal Truth Table

Instruction	Opcode	Funct7	Funct3	ALUctrl	ALUsrc	REGwr	MEMwr	MEMrd	MEMtoREG	PCsrc
ADD	0110011	0000000	000	000	0	1	0	0	0	0
SUB	0110011	0100000	000	001	0	1	0	0	0	0
AND	0110011	0000000	111	010	0	1	0	0	0	0
OR	0110011	0000000	110	011	0	1	0	0	0	0
XOR	0110011	0000000	100	100	0	1	0	0	0	0
SLT	0110011	0000000	010	101	0	1	0	0	0	0
SLL	0110011	0000000	001	110	0	1	0	0	0	0
SRL	0110011	0000000	101	111	0	1	0	0	0	0
ADDI	0010011	NA	000	000	1	1	0	0	0	0
ANDI	0010011	NA	111	010	1	1	0	0	0	0
ORI	0010011	NA	110	011	1	1	0	0	0	0
XORI	0010011	NA	100	100	1	1	0	0	0	0
SLTI	0010011	NA	010	101	1	1	0	0	0	0
SLLI	0010011	NA	001	110	1	1	0	0	0	0
SRLI	0010011	NA	101	111	1	1	0	0	0	0
LW	0000011	NA	010	000	1	1	0	1	1	0
SW	0100011	NA	010	000	1	0	1	0	0	0
BEQ	1100011	NA	000	001	0	0	0	0	0	1*
BNE	1100011	NA	001	001	0	0	0	0	0	1*

* BEQ: PCsrc=1 when Zero=1 | BNE: PCsrc=1 when Zero=0

3. ALU Decoder (2-Level Control)

Type	ALUop	Funct7[5]	Funct3	ALU Operation	Notes
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R-type	00	0	000	ADD (000)	
R-type	00	1	000	SUB (001)	
R-type	00	0	111	AND (010)	
R-type	00	0	110	OR (011)	
R-type	00	0	100	XOR (100)	
R-type	00	0	010	SLT (101)	
R-type	00	0	001	SLL (110)	
R-type	00	0	101	SRL (111)	
I-type	01	X	funct3	Same as R-type per funct3	ALUsrc=1, use imm
LW/SW	10	X	010	ADD (000)	Address = rs1 + imm
BEQ	11	X	000	SUB (001)	Zero flag -> PCsrc
BNE	11	X	001	SUB (001)	~Zero flag -> PCsrc

4. Main Decoder (Opcode to Control Signals)

Opcode	Type	ALUop	ALUsrc	IMMsrc	REGwr	MEMwr	MEMrd	MEMtoREG	Bop
0110011	R-type	00	0	X (none)	1	0	0	0	00
0010011	I-type (ALU)	01	1	00 (I)	1	0	0	0	00
0000011	I-type (Load)	10	1	00 (I)	1	0	1	1	00
0100011	S-type	10	1	11 (S)	0	1	0	0	00
1100011	B-type	11	0	01 (B)	0	0	0	0	10/01

IMMsrc: 00=I-immediate | 01=B-immediate | 11=S-immediate | Bop: 10=BEQ | 01=BNE

5. Instruction Descriptions

Group	Instruction	Format	Description / Dataflow
R-type	ADD	R	rs1 + rs2 -> rd RegFile(rs1,rs2) -> ALU(ADD) -> RegFile(rd)
	SUB	R	rs1 - rs2 -> rd ALU(SUB) -> RegFile(rd)
	AND	R	rs1 AND rs2 -> rd Bitwise AND
	OR	R	rs1 OR rs2 -> rd Bitwise OR
	XOR	R	rs1 XOR rs2 -> rd Bitwise XOR
	SLT	R	rd = 1 if rs1 < rs2 (signed), else 0
	SLL	R	rs1 << rs2[4:0] -> rd Logical left shift, zeros inserted
	SRL	R	rs1 >> rs2[4:0] -> rd Logical right shift, zeros inserted
I-type (ALU)	ADDI	I	rs1 + sign_ext(imm[11:0]) -> rd
	ANDI	I	rs1 AND sign_ext(imm) -> rd
	ORI	I	rs1 OR sign_ext(imm) -> rd

	XORI	I	rs1 XOR sign_ext(imm) -> rd
	SLTI	I	rd = 1 if rs1 < sign_ext(imm), else 0
	SLLI	I	rs1 << imm[4:0] -> rd imm[11:5] must be 0
	SRLI	I	rs1 >> imm[4:0] -> rd imm[11:5] must be 0
Memory	LW	I	Addr = rs1 + sign_ext(imm) -> Mem[Addr] -> rd
	SW	S	Addr = rs1 + sign_ext(imm) ; Mem[Addr] = rs2
Control Flow	BEQ	B	if (rs1 == rs2): PC = PC + sign_ext(imm<<1) Zero flag -> PCsrc
	BNE	B	if (rs1 != rs2): PC = PC + sign_ext(imm<<1) ~Zero flag -> PCsrc